D-exp-SP2: Topological Optimality Criterion for Coherent Photon-to-Electron Conversion

Projective Dynamic Logo Framework — Exploratory Document

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Epistemic Status

This is an **exploratory document**. It is not part of the PDL core corpus (D01–D53). Its results are classified as follows:

- **Theorems**: Theorem 3.1 (PDL-SP2) and its corollaries — proved analytically and verified exhaustively.
- **Observations**: Rule A ($k_{\text{avg}} = 4$, $\Delta\text{EN} > 1.0 \Rightarrow$ direct gap) — verified on 15 materials, not a theorem.
- **Conjectures**: PDL-H (Conjecture 6.1) — verified exhaustively on 2^{24} cases, not yet proved analytically.
- **Predictions**: Falsifiable experimental predictions (Section 7) — no free parameters.

No material was presupposed optimal prior to computation.

Abstract

We derive a topological optimality criterion for coherent photon-to-electron conversion from the axioms C1–C4 of the Projective Dynamic Logo (PDL) framework. The central result (Theorem PDL-SP2) establishes that the minimal geometric leakage $\varepsilon_{\text{geom}} = 0$ is achievable on a bond graph G if and only if G is bipartite — recovering the Harary balance theorem (1953) as a corollary of C1–C4 without presupposition. A four-criterion composite score derived from PDL selects six optimal inorganic candidates from 21 families: BP, AlAs, GaP, AlP, GaN, and SiC. GaN and BP are identified as priority candidates with direct gaps of 1.73 eV and 1.24 eV respectively. An extension to hierarchical graphs (Conjecture PDL-H) shows that global bipartition is the sole determining condition — local graph structure is irrelevant for $\varepsilon_{\text{geom}}(H) = 0$. This result is verified exhaustively on $2^{24} = 16\,777\,216$ sign assignments for the case $H(C_5, C_4)$. Biological connection: the Mg–Mg transfer network of Photosystem II is bipartite, consistent with PDL-H, while individual chlorophyll molecules are not — a two-scale architecture that PDL explains naturally. Three falsifiable experimental predictions are formulated, testable by femtosecond pump-probe spectroscopy without free parameters.

Contents

1 Introduction and Motivation

The conversion of photons to free electrons with near-unity quantum efficiency is the central challenge of photovoltaic science and the defining achievement of biological photosynthesis. Photosystem II achieves charge separation in under 3 ps with a quantum yield approaching 100% [1]. Current best photovoltaic materials operate far below this limit, and no theoretical framework has derived from first principles which structural features make a material capable of achieving such efficiency.

The Projective Dynamic Logo (PDL) framework [2] derives fundamental physical structures from four combinatorial axioms (C1–C4) on finite signed graphs, without presupposing spacetime, particles, or fields. Its central object is the graph K_4, identified as the minimal admissible closure. Document D49 of the PDL corpus derives the London equation from C1–C4 via the metric identity $\nu = \frac{1}{4}\|\psi_1 - T^k\psi_1\|^2$, characterising coherent transport as a norm of the difference between a state and its image under K_4 pulsation. Document D43 defines the geometric leakage parameter $\varepsilon_{\text{geom}}(G, v)$ as the fraction of cycles of length ≤ 6 incident to node v with negative sign product.

This document applies these PDL tools to a concrete problem: identifying, from first principles, which materials and material architectures are topologically optimal for coherent photon-to-electron conversion. The approach is strictly deductive — no material is presupposed optimal, and all candidate selection follows from the axioms.

2 PDL Ingredients

We recall the four PDL results used throughout this document. All are theorems of the core corpus; their proofs are not reproduced here.

Definition 2.1: Geometric Leakage (D43)

Let $G = (V, E, s)$ be a finite signed graph with $s : E \to \{+1, -1\}$. For a node $v \in V$, define

$$\varepsilon_{\text{geom}}(G, s, v) = \frac{\left|\{C \in \mathcal{C}_{\leq 6}(v) : \prod_{e \in C} s(e) = -1\}\right|}{|\mathcal{C}_{\leq 6}(v)|}$$

where $\mathcal{C}_{\leq 6}(v)$ denotes the set of simple cycles of length ≤ 6 incident to v. The global leakage is $\varepsilon_{\text{geom}}(G, s) = \frac{1}{|V|}\sum_v \varepsilon_{\text{geom}}(G, s, v)$.

Definition 2.2: Bipartite Graph

A graph $G = (V, E)$ is *bipartite* if V can be partitioned into two disjoint sets A and B such that every edge connects a vertex in A to a vertex in B. Equivalently, G contains no cycle of odd length.

The four PDL results used here are:

P1. Minimal closure (D16a): K_4 is the smallest admissible graph satisfying C1–C4.

P2. Geometric leakage (D43): $\varepsilon_{\text{geom}}(p) = 329/10087$ and $\varepsilon_{\text{geom}}(n) = 468/9960$ are unconditional theorems of C1–C4.

P3. Coherent transport (D49): The London equation is derived from C1–C4 via $\nu = \frac{1}{4}\|\psi_1 - T^k\psi_1\|^2$.

P4. Connectivity (C1): Every PDL-admissible graph is finite and connected. This forces the number of connected components $c = 1$, and hence $|\text{eps}_0| = 2^{N-1}$ (see below).

3 Theorem PDL-SP2: Bipartition and Zero Leakage

Theorem 3.1: PDL-SP2 — Bipartition and Zero Leakage

Let G be a connected graph with N nodes and $c = 1$ connected component. A sign assignment $s : E \to \{+1, -1\}$ satisfies $\varepsilon_{\text{geom}}(G, s) = 0$ if and only if there exists a colouring $f : V \to \{+1, -1\}$ such that $s(u, v) = f(u) \cdot f(v)$ for every edge $(u, v) \in E$. Equivalently: $\varepsilon_{\text{geom}}(G, s) = 0$ is achievable if and only if G is bipartite.

Proof sketch. **Part A** (every s_f satisfies $\varepsilon_{\text{geom}} = 0$): For any cycle $(v_1, \ldots, v_k, v_1)$ and any colouring f,

$$\prod_{i=1}^{k} s_f(v_i, v_{i+1}) = \prod_{i=1}^{k} f(v_i) \cdot f(v_{i+1}) = \prod_{i=1}^{k} f(v_i)^2 = +1,$$

since each $f(v_i) \in \{+1, -1\}$ appears exactly twice. Hence every cycle has positive sign product and $\varepsilon_{\text{geom}} = 0$.

Part B (reconstruction): Conversely, given s with $\varepsilon_{\text{geom}} = 0$, fix a root node r with $f(r) = +1$. For any other node v, choose a path $P(r \to v)$ and define $f(v) = \prod_{e \in P} s(e)$. Since every cycle has product $+1$, this definition is path-independent (connectivity is essential here). One verifies $s(u, v) = f(u) \cdot f(v)$ for every edge. □

Connection to Harary (1953): Theorem 3.1 recovers the Harary balance theorem [3] as a consequence of C1–C4, without presupposing it.

Theorem 3.2: Counting Optimal Assignments

For a connected graph G with N nodes, M edges, and k-regular with $k = 3$ (3-regular, $M = 3N/2$):

$$|\{\text{sign assignments with } \varepsilon_{\text{geom}} = 0\}| = 2^{N-1}, \qquad \text{fraction} = \frac{2^{N-1}}{2^{3N/2}} = 2^{-N/2-1}.$$

The bijection $f \mapsto s_f$ is 2-to-1 (since f and $-f$ give the same assignment), accounting for the factor of $1/2$.

Exhaustive verification: Both theorems were verified exhaustively for all 3-regular graphs with $N = 4, 6, 8, 10$ nodes (7 distinct graphs, all sign assignments enumerated), confirming zero exceptions. The reconstruction algorithm (Part B) was verified on all optimal assignments for each graph, with zero failures.

For non-connected graphs with c components, the counting formula generalises to 2^{N-c}, verified on the disjoint union of two K_4 graphs ($N = 8$, $c = 2$, $2^6 = 64$ optimal assignments confirmed).

4 Optimality Criterion for Photon-to-Electron Conversion

Theorem 3.1 establishes bipartition as the necessary and sufficient condition for zero geometric leakage. For application to photovoltaic materials, three additional conditions are derived from physical considerations anchored in PDL.

Definition 4.1: PDL-SP2 Composite Score

A material with bond graph G, band gap E_g, and bipartition groups A, B receives a score $\sigma \in \{0, 1, 2, 3, 4\}$:

$$\sigma = \mathbf{1}[G \text{ bipartite}] + \mathbf{1}[G \text{ connected}] + \mathbf{1}[1.1 \leq E_g \leq 2.5 \text{ eV}] + \mathbf{1}[\text{elements}(A) \neq \text{elements}(B)].$$

Criterion 1 is the PDL theorem; Criterion 3 targets the solar spectrum; Criterion 4 requires chemical heterogeneity (symmetry breaking) to create a preferred direction of charge transfer.

Rule A (Observation, not theorem): Among 15 materials tested, the rule $k_{\text{avg}} = 4$ and $\Delta\text{EN} > 1.0 \Rightarrow$ direct gap holds for 4/4 materials satisfying both conditions (GaN, AlN, InN, ZnO). The sole exception at $\Delta\text{EN} > 1.0$ is SnO_2 ($k_{\text{avg}} = 2.67$, rutile structure), correctly excluded by the k_{avg} condition. This observation is not a PDL theorem; DFT validation is required.

5 Material Screening: 21 Candidates from Four Families

Bond graphs were constructed from crystallographic structures retrieved from the Materials Project [4] using covalent-radius cutoffs (Alvarez 2008 [5]). Each structure was retrieved dynamically by chemical formula (most stable phase, $E_{\text{hull}} = 0$), with systematic element verification to exclude database artefacts.

5.1 Families Tested

Four families were selected from PDL optimality conditions, without presupposing any candidate:

- **F1**: III-V nitrides and carbides (GaN, AlN, InN, BN, GaP, InP, AlP, SiC)
- **F2**: Chalcopyrites and sulfo-salts ($CuInSe_2$, $CuGaSe_2$, $CuInS_2$, $AgInSe_2$, $ZnSnP_2$, $CuGaS_2$)
- **F3**: Graphene analogues (BN, BP, AlAs, InAs)
- **F4**: Ternary oxides and spinels (ZnO, $CuAlO_2$, $CuGaO_2$, $ZnGa_2O_4$, $MgAl_2O_4$, Cu_2O, SnO_2)

5.2 Results

Priority candidates: GaN (score 4/4, direct gap 1.73 eV, $\Delta\text{EN} = 1.23$, Rule A satisfied) and BP (score 4/4, gap 1.24 eV near the Shockley–Queisser optimum of 1.34 eV, largely unexplored for photovoltaics).

Table 1: PDL-SP2 screening results. Score 4/4 = optimal. Gap type: D = direct, I = indirect (experimental literature). Bipartition groups listed for score-4 candidates only.

Material	Score	E_g (eV)	Type	Sym. group	Group A	Group B
BP	4/4	1.24	I	F-43m	{B}	{P}
AlAs	4/4	1.50	I	F-43m	{Al}	{As}
GaP	4/4	1.60	I	F-43m	{Ga}	{P}
AlP	4/4	1.63	I	F-43m	{Al}	{P}
GaN	4/4	1.73	D	$P6_3mc$	{Ga}	{N}
SiC	4/4	1.84	I	R3m	{Si}	{C}
InP	3/4	0.45	D	F-43m	—	—
ZnO	3/4	0.72	D	$P6_3mc$	—	—
AlN	3/4	4.05	D	$P6_3mc$	—	—
$ZnSnP_2$	3/4	0.70	D	I-42d	—	—
SnO_2	3/4	0.65	I	$P4_2/mnm$	—	—
Cu_2O	0/4	0.51	—	Pn-3m	non-bipartite	
$MgAl_2O_4$	0/4	5.11	—	Fd-3m	non-bipartite	

6 Conjecture PDL-H: Hierarchical Graphs

Biological photosynthesis operates at two topological scales simultaneously: individual chlorophyll molecules contain a five-membered ring (cyclopentanone, girth = 5, non-bipartite), yet the Mg–Mg transfer network between chlorophylls is bipartite (verified on structure 3ARC of Photosystem II, Protein Data Bank [6]). This two-scale architecture motivates the following extension of PDL-SP2.

Definition 6.1: Hierarchical Graph

A hierarchical graph $H = (G_{\text{loc}}, G_{\text{glob}}, \phi)$ is defined by:

- G_{loc}: local graph (intra-chromophore), any structure;
- G_{glob}: global graph (inter-chromophore network);
- $\phi \in V(G_{\text{loc}})$: connection node (analogue of Mg).

The total graph H places one copy of G_{loc} on each node of G_{glob}, connecting adjacent copies via ϕ.

Conjecture 6.1: PDL-H — Global Bipartition Dominates

For any hierarchical graph $H = (G_{\text{loc}}, G_{\text{glob}}, \phi)$ with G_{loc} connected:

$$\varepsilon_{\text{geom}}(H) = 0 \text{ is achievable} \iff G_{\text{glob}} \text{ is bipartite},$$

independently of the structure of G_{loc} (its girth, bipartition status, or size).

Exhaustive verification: The case $H(C_5, C_4)$ — pentagonal local graph (girth = 5, non-bipartite) with square global graph (bipartite) — was verified exhaustively over all $2^{24} = 16\,777\,216$ sign assignments. Result: $\varepsilon_{\text{geom}} = 0$ is achievable, with $524\,288 = 2^{19}$

optimal assignments. This is consistent with the counting formula 2^{N-1} (here $N = 20$, 2^{19} pairs under the symmetry $f \sim -f$).

Additional verification: cycles C_n for $n = 3, \ldots, 12$ as local graphs, combined with three bipartite global graphs (P4, C4, C6). In all computable cases (exact enumeration or greedy heuristic), $\varepsilon_{\mathrm{geom}}(H) = 0$ was achieved when G_{glob} is bipartite.

Biological implication: Conjecture PDL-H explains why Photosystem II operates efficiently despite individual chlorophylls being non-bipartite. The bipartite Mg–Mg network is sufficient for $\varepsilon_{\mathrm{geom}}(H) = 0$ at the system level.

Design implication: For artificial photovoltaic systems, PDL-H implies that the local chromophore structure is topologically irrelevant — only the global network topology must be bipartite. This substantially relaxes the material design constraints identified by PDL-SP2 alone.

7 Falsifiable Predictions

Prediction 7.1: Charge Separation Hierarchy

Among the score-4/4 candidates, the charge separation time τ_{cs} satisfies the hierarchy

$$\tau_{\mathrm{cs}}(\mathrm{GaN}) < \tau_{\mathrm{cs}}(\mathrm{BP}) < \tau_{\mathrm{cs}}(\mathrm{SiC}) < \tau_{\mathrm{cs}}(\mathrm{GaP}),$$

with $\tau_{\mathrm{cs}}(\mathrm{GaN}) < 3$ ps, testable by femtosecond transient absorption spectroscopy under identical excitation conditions. No free parameters. Rationale: GaN satisfies Rule A (direct gap, $k = 4$, $\Delta\mathrm{EN} > 1.0$); the hierarchy follows from decreasing $\Delta\mathrm{EN}$ and direct-to-indirect gap transition.

Prediction 7.2: Hierarchical Photovoltaic Architecture

A van der Waals heterostructure of two score-4/4 materials (e.g. GaN/BP or GaN/SiC) with a bipartite inter-layer coupling network will exhibit charge separation times comparable to Photosystem II ($\tau_{\mathrm{cs}} < 3$ ps), by Conjecture PDL-H. The external quantum efficiency will exceed that of either material in isolation. Testable by EQE measurement and pump-probe spectroscopy.

Prediction 7.3: Non-Bipartite Disqualification

Materials with non-bipartite bond graphs (Cu_2O, $MgAl_2O_4$) will exhibit structurally longer charge separation times than bipartite candidates under identical geometry and excitation, reflecting the irreducible leakage $\varepsilon_{\mathrm{geom}} > 0$ imposed by odd cycles in the bond graph. Testable by comparative femtosecond spectroscopy.

Table 2: Epistemic status of all results in this document.

Result	Status	Evidence
Theorem PDL-SP2 ($\varepsilon_{\mathrm{geom}} = 0 \Leftrightarrow$ bipartite)	Theorem	Analytic proof + exhaustive verification, 7 graphs
Counting formula 2^{N-1}	Theorem	Exhaustive, $N = 4, 6, 8, 10$
Harary (1953) recovered from C1–C4	Theorem	Analytic
Score-4/4 candidates (GaN, BP, AlAs, GaP, AlP, SiC)	Observation	Materials Project screening, 21 materials
Rule A ($k = 4$, $\Delta\mathrm{EN} > 1.0 \Rightarrow$ direct gap)	Observation	15 materials, 4/4 correct, 1 exception (SnO_2)
Conjecture PDL-H	Conjecture	Exhaustive 2^{24}, heuristic $n = 3..12$
Predictions P1–P3	Prediction	Derived from PDL-SP2 + PDL-H, no free parameters

8 Epistemic Table

9 Open Problems

Open Problem 9.1: Analytic Proof of PDL-H

Prove Conjecture 6.1 analytically for all connected hierarchical graphs $H = (G_{\mathrm{loc}}, G_{\mathrm{glob}}, \phi)$. The exhaustive verification establishes the result for $H(C_5, C_4)$ and heuristic evidence covers C_n for $n = 3, \ldots, 12$, but a general proof requires an argument independent of the size of G_{loc}.

Open Problem 9.2: Criterion for the Ambiguous Region

Rule A fails for materials with $\Delta\mathrm{EN} < 1.0$: GaAs (direct) and GaP (indirect) have nearly identical $\Delta\mathrm{EN}$ (0.37 vs. 0.38) and cannot be distinguished by topological means alone. Identify an additional PDL-derivable criterion — possibly related to orbital character at band edges — that resolves this ambiguity.

Open Problem 9.3: Extension to Molecular Frameworks

Apply PDL-H to covalent organic frameworks (COF) based on metalloporphyrins. These systems have non-bipartite local chromophores (girth = 5) organised in hexagonal 2D networks (bipartite globally). PDL-H predicts $\varepsilon_{\mathrm{geom}}(H) = 0$; this should be verified on crystallographic structures from the Cambridge Structural Database, and compared with measured quantum yields.

10 Computational Resources

All numerical verifications were performed in Google Colaboratory using Python 3.10 with NetworkX 3.x, pymatgen, and the Materials Project API (mp-api). Crystallographic data retrieved from the Materials Project [4]. Biological structure 3ARC from the RCSB Protein Data Bank [6], parsed with Biopython. All computations use exact arithmetic (Python `Fraction` class) where applicable. The following Colab notebooks constitute the complete computational trace of this document:

NB1. `PDL_OP_SP2_derivation.ipynb` — Theorem PDL-SP2, bijection verification, counting formula, connectivity necessity.

NB2. `PDL_OP_SP2_proof.ipynb` — Analytic proof verification (Part A telescoping, Part B BFS reconstruction, non-connected case).

NB3. `PDL_OP_SP2_clean.ipynb` — Material screening (7 candidates, covalent cutoff, bipartition, dynamic formula lookup).

NB4. `PDL_OP_SP2_extended.ipynb` — Extended screening (21 materials, 4 families, composite score 0–4).

NB5. `PDL_OP_SP2_stepA.ipynb` — Rule A derivation (ΔEN vs. direct/indirect gap, 15 materials).

NB6. `PDL_OP_SP2_stepA_corrected.ipynb` — Corrected supercell analysis (girth and k with 2×2×2 cells).

NB7. `PDL_OP_SP2_stepB.ipynb` — Biological verification on Photosystem II structure 3ARC (chlorophyll, Mg–Mg network).

NB8. `PDL_OP_SP2_hierarchical.ipynb` — Hierarchical graph analysis (Verification 1, all combinations).

NB9. `PDL_OP_SP2_verification2.ipynb` — Verification 2 (C_n for $n = 3, \ldots, 12$, pair/odd frontier).

NB10. `PDL_OP_SP2_exhaustive_C5C4.ipynb` — Exhaustive verification $H(C_5, C_4)$, 2^{24} assignments, 20 minutes computation time, $\varepsilon_{\text{geom}} = 0$ confirmed.

All notebooks are archived with this document on Zenodo (community: `pdl-framework`).

References

[1] G. R. Fleming, J. L. Martin, and J. Breton. Rates of primary electron transfer in photosynthetic reaction centres and their mechanistic implications. *Nature*, 333:190–192, 1988.

[2] C. Laubscher. PDL framework — master document v20 (D01–D53). Zenodo community: `pdl-framework`, 2026. https://zenodo.org/communities/pdl-framework.

[3] F. Harary. On the notion of balance of a signed graph. *Michigan Mathematical Journal*, 2(2):143–146, 1953.

[4] A. Jain, S. P. Ong, G. Hautier, W. Chen, W. D. Richards, S. Dacek, S. Cholia, D. Gunter, D. Skinner, G. Ceder, and K. A. Persson. Commentary: The materials project: A materials genome approach to accelerating materials innovation. *APL Materials*, 1:011002, 2013.

[5] S. Alvarez. A cartography of the van der Waals territories. *Dalton Transactions*, pages 2832–2838, 2008.

[6] Y. Umena, K. Kawakami, J.-R. Shen, and N. Kamiya. Crystal structure of oxygen-evolving photosystem II at a resolution of 1.9 å. *Nature*, 473:55–60, 2011.